

JKBOSE 2026

CLASS 11TH

Bhalcharya
on
TOP

STRUCTURE OF ATOM

LECTURE-6

CHEMISTRY

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Topics To Covered

- 1 De-Broglie Relationship (Most Imp)
- 2 HEISENBERG'S UNCERTAINTY PRINCIPLE. (Most Imp).
- 3 QUANTUM NUMBERS (Introduction).

REVISION.

chapter
end

1) Dual nature of Radiation and matter.

↓
Light → Wave
→ Photon.

Matter → Wave
→ Particle

$$\lambda = \frac{h}{mv}$$

DE-BROGLIE RELATIONSHIP

According to Einstein mass energy relationship.

$$E = mc^2 \longrightarrow \textcircled{1} \quad \begin{array}{l} m = \text{mass of particle} \\ c = \text{velocity of light} \end{array}$$

According to Planck's Quantum Theory

$$E = h\nu \longrightarrow \textcircled{2} \quad \begin{array}{l} h = \text{planck's constant} \\ 6.626 \times 10^{-34} \text{ Js} \\ \text{OR} \\ 6.626 \times 10^{-34} \text{ kgm}^2\text{s}^{-1} \end{array}$$

From 1 & 2.

$$h\nu = mc^2$$

$$\rightarrow \cancel{h} \frac{c}{\lambda} = m \cancel{c^2}$$

$$\Rightarrow \frac{h}{\lambda} = mc$$

$$\nu = \text{frequency} \\ (\nu = \frac{c}{\lambda})$$



Significance of de-broglie relationship

De-broglie relationship significant for only microscopic particle i.e. λ can be easily calculated for microscopic particle.

For macroscopic particle, mass is very high & mass is inversely proportional to the wavelength hence ' λ ' for macroscopic particle comes out to be very small, can not be calculated easily.

$$\frac{h}{\lambda} = mc$$

$$\text{or } \frac{h}{mc} = \lambda$$

$$\text{or } \lambda = \frac{h}{mc}$$

If we replace 'c' with 'v' (velocity of a particle)

$$\lambda = \frac{h}{mv}$$

$$\text{or } \lambda = \frac{h}{p}$$

x

Heisenbergs Uncertainty Principle.

It was given by Werner Heisenberg.

It states that "it is impossible to calculate exact position and momentum of a microscopic particle such as e^- ."

Mathematically.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

Δx = uncertainty in position

Δp = Uncertainty in momentum

h = Planck constant

6.626×10^{-34} Js.

OR 6.626×10^{-34} kgm²s⁻¹



Heisenberg.

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\text{OR } \Delta x \cdot m\Delta v \geq \frac{h}{4\pi}$$

$$\text{OR } \Delta x \cdot \Delta v \geq \frac{h}{4\pi m}$$

$$(p = mv)$$

★

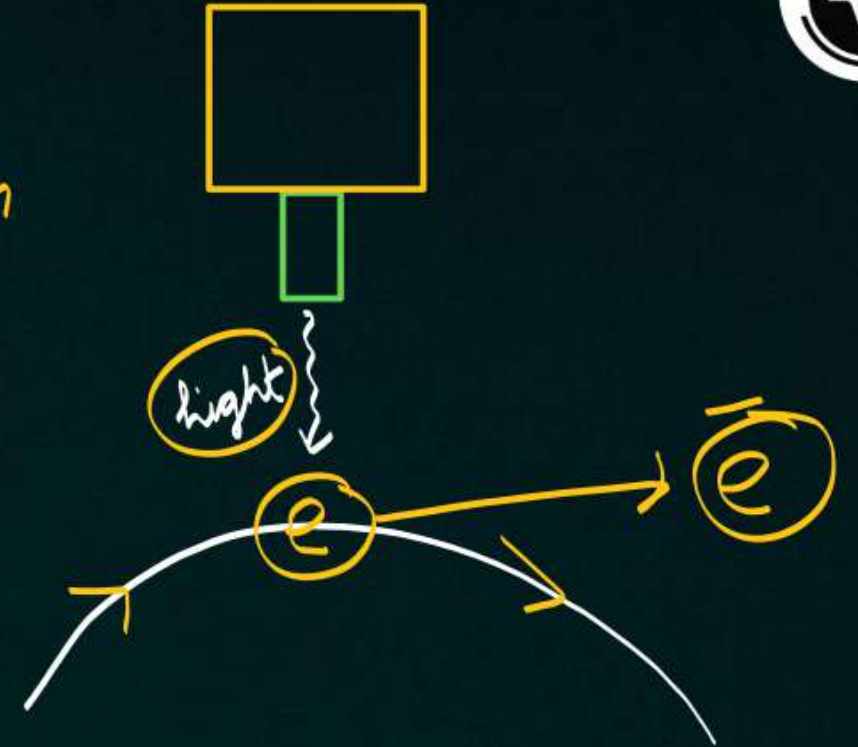
Bright light of high intensity
of high energy

REASON $\Rightarrow$

To determine the position we need light of high intensity (Bright light), but bright light means light of high energy.

and when light of high energy strikes the e^- then its momentum will get uncertain

And if we use light of less intensity, the position will become uncertain.



QUESTION



~~-2/4~~

What will be the wavelength of a ball of mass 0.1 kg moving with a velocity of 10 ms^{-1} ?

Sol. Given:→

$$m = 0.1 \text{ kg}$$

$$v = 10 \text{ ms}^{-1}$$

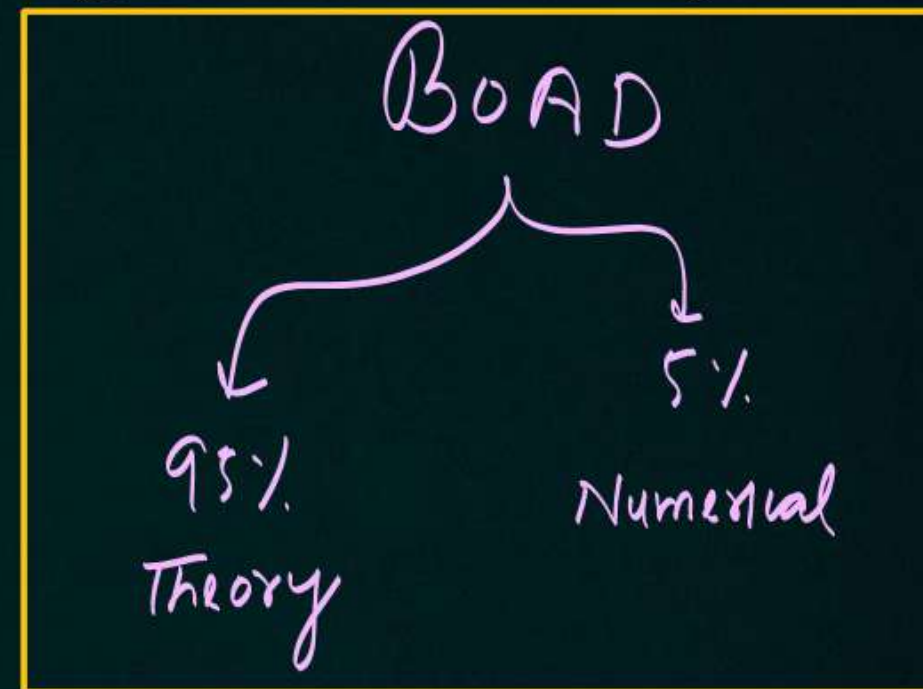
$$\lambda = ?$$

$$\lambda = \frac{h}{mv}$$

$$= \frac{6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}}{0.1 \text{ kg} \times 10 \text{ m s}^{-1}}$$

$$= \frac{6.626 \times 10^{-34} \text{ m}}{\frac{1}{10} \times 10}$$

$$= 6.626 \times 10^{-34} \text{ m} \checkmark$$



QUESTION

Δx

$$0.1 = \frac{1}{10} = 10^{-1}$$



The uncertainty in the position of an electron is 0.1 nm. What is the uncertainty in its momentum?

1 $5.27 \times 10^{-25} \text{ kg m/s}$

2 $6.62 \times 10^{-25} \text{ kg m/s}$

3 $1.06 \times 10^{-25} \text{ kg m/s}$

4 $3.29 \times 10^{-25} \text{ kg m/s}$

$$\Delta x \cdot \Delta p = \frac{h}{4\pi}$$

$$\Delta p = \frac{h}{4\pi \Delta x} = \frac{6.626 \times 10^{-34} \text{ kg m}^2 \text{ s}^{-1}}{4 \times 3.141 \times 0.1 \text{ nm}}$$

$$= \frac{6.626 \times 10^{-34}}{4 \times 3.141 \times 0.1 \times 10^{-9}}$$

$$= \frac{6.626 \times 10^{-24}}{4 \times 3.141 \times 10^{-1} \times 10^{-9}}$$

$$0.527 \times 10^{-24}$$

$$1 \text{ nm} = 10^{-9} \text{ m}$$

$$1 \text{ pm} =$$

$$1 \text{ \AA} =$$

Significance of Uncertainty Principle

- This principle means that electrons and similar tiny particles don't follow a fixed path. We can't say exactly where they are or how they move we can only talk about the chances of where they might be. *(concept of orbit was discarded)*.
- However, the effect of this principle is significant only in case of **microscopic objects** and is negligible in case of macroscopic object.

QUESTION

Which of the following is responsible to rule out the existence of definite paths or trajectories of electrons. (orbit).

- 1 Pauli's exclusion principle
- 2 Heisenberg's uncertainty principle
- 3 Hund's rule of maximum multiplicity
- 4 Aufbau principle

QUESTION

If travelling at same speeds, which of the following matter waves have the shortest wavelength?

$$\lambda \propto \frac{1}{m} \quad \downarrow \uparrow m$$

1 Electron ✓

2 Alpha particle (He^{2+}) ✓ $\begin{matrix} 4 \\ 2 \end{matrix} \text{He}^{2+}$

3 Neutron ✓

4 Proton ✓

QUESTION

The uncertainty principle is significant for:

- 1 Macroscopic objects
- 2 Microscopic objects
- 3 Both macroscopic and microscopic objects
- 4 Objects with definite paths

Moongfali

- ①(Abbas)²
- ②(Aleem)'
- ③(Musaa.)'

$$M.M = (2 \times \underline{VD})$$

vapour density

{ REVISION }



Thank
You